

Cole Schreiner

Robotics Engineer

🔗 Portfolio: ColeSchreiner.com [in linkedin.com/in/cole-schreiner](https://www.linkedin.com/in/cole-schreiner) ✉️ ColeRobotEngineer@gmail.com

Education

University of California, Santa Cruz

Expected June 2026

Robotics Concentration ~ Graduated with Honors

- Master of Science in Electrical and Computer Engineering
- Bachelor of Science in Robotics Engineering
- Minor in Electrical Engineering

Technical Skills

Languages: C, C++, Java, Python, Verilog, MATLAB

Embedded and Control: ESP32, Raspberry Pi, PIC32, Arduino, FreeRTOS, PID control, finite state machines, real-time sensing, data logging

Design and Fabrication: SolidWorks, OnShape, AutoDesk, 3D printing, laser cutting, prototyping, electronics assembly, instrumentation and oscilloscope-based debugging

Software and Analysis: Git, Vivado, LTSpice, PyTorch, GUI development, web-based interfaces, signal processing, peak detection, normalization, and model training

Circuit and Systems: Analog filters, op-amps, power regulation, motor drivers, feedback control, RL/RLC analysis, frequency response, and sensor signal conditioning

Engineering Areas: Robotics, mechatronics, kinematics, microcontroller system design, VLSI, scientific instrumentation, machine learning, and imitation learning, prompt engineering

Work Experience

University of California, Santa Cruz

Spring 2026

Graduate Teaching Assistant - Mechatronics

- Selected to support undergraduate mechatronics instruction through hands-on laboratory assignments spanning electrical systems, mechanical integration, embedded software, analog circuits, and motor control.
- Mentor student teams through subsystem debugging, design iteration, and system integration as they build toward the course's final robot competition.
- Translate complex sensing, actuation, control, and fabrication concepts into practical lab guidance that helps students move from theory to a working robotic platform.

Major Engineering Projects

UAV Artificial Intelligence Autopilot and Simulation [\[Project Link\]](#)

February, 2026

Neural Flight Control and Behavioral Cloning

- Reframed a traditional small-scale UAV autopilot problem by replacing successive loop-closure control logic with a recurrent neural network trained through behavioral cloning.
- Built a PyTorch inference pipeline that mapped 15 flight-state inputs, including attitude, body-frame velocity, and airflow features, to continuous throttle, aileron, elevator, and rudder commands.
- Integrated the learned controller into a university flight-simulation framework to evaluate autonomous path following, closed-loop stability, and disturbance rejection under simulated wind conditions.
- Applied data normalization and mean-squared-error training against expert teacher demonstrations to validate whether a single learned controller could manage coupled flight dynamics in real time.

Capstone - Wet-Dry Cycler (RNA Replicator) - [\[Project Link\]](#)

2024 - 2025

Project Director - Origins of Life Simulation Device

- Led a multidisciplinary capstone team in the design of an automated wet-dry cycling instrument for origins-of-life and RNA/DNA polymerization research, translating biochemistry research needs into engineering requirements and testable system specifications.
- Developed an ESP32-based control system with programmable cycle timing, environmental sensing, safety interlocks, and closed-loop temperature and humidity regulation for repeatable laboratory protocols.
- Designed a modular platform with multiple sample chambers, integrated data logging, remote monitoring, and web-based user interaction to support longer unattended experiments and protocol reconfiguration.
- Managed technical integration, documentation, validation testing, and stakeholder communication with researchers while balancing mechanical design, embedded software, and experimental reliability constraints.

Professor Piano - Wearable Piano Teaching Glove - [\[Project Link\]](#)

March, 2025

Wearable Sensing and Interactive Feedback System

- Designed a wearable piano instruction glove that combined piezoelectric sensors, a 9-axis IMU, and an ESP32 to capture finger force, hand motion, and playing posture in real time.
- Implemented signal-processing pipelines for peak detection, filtering, calibration, and movement interpretation to improve the reliability of finger press and orientation measurements across users.
- Integrated I2S audio playback to provide immediate auditory feedback and support an interactive training experience focused on technique reinforcement and muscle-memory development.
- Contributed to the system concept across electronics, embedded software, and ergonomic wearable design, including flexible integration considerations for comfort, repeatability, and extended use.

Mechatronics Robot Competition - [\[Project Link\]](#)

2024

Team Integration Director - Autonomous Competition Robot

- Directed subsystem integration for a five-week autonomous robot competition project, coordinating mechanical, electrical, and embedded software development under an aggressive build and test schedule.
- Built the robot around a finite state machine architecture with sensor-driven behaviors for navigation, task execution, and fault handling, supporting deterministic operation and rapid iteration during development.
- Designed custom electrical subsystems including analog filters, motor-control electronics, power distribution, and sensor signal-conditioning circuits to improve robustness in a noisy competition environment.
- Managed interface definitions, team timelines, integrated testing, and system troubleshooting as the robot progressed from concept to validated competition platform against 20 teams.

Palos Verdes Summer Tech Internship [\[Project Link\]](#)

Summer 2021

Senior Engineering Intern - Canine Gait Analysis Wearable Project

- Directed the development of a wearable diagnostic device intended to help veterinarians detect abnormal gait patterns in dogs earlier and at lower cost than conventional screening workflows.
- Integrated an Arduino Nano, MPU-9250 inertial sensing, battery regulation, MicroSD logging, and harness-mounted packaging into a portable prototype capable of collecting multi-axis motion data during live canine walking trials.
- Conducted data collection and test execution with healthy dog subjects to build baseline datasets for a machine-learning pipeline aimed at distinguishing healthy and abnormal stride behavior.
- Helped connect recorded motion data to downstream visualization and analysis workflows designed to identify likely gait irregularities and improve veterinarian insight.